Amazon Customer Behavior Analysis Report

1. Introduction

This project focuses on analyzing Amazon customer behavior using Python. The aim is to understand purchasing patterns, customer preferences, and product relationships to improve recommendations and business strategies.

2. Objectives

- Analyze customer purchasing behavior
- Identify popular product categories
- Perform customer segmentation
- Apply Market Basket Analysis
- Generate actionable business insights

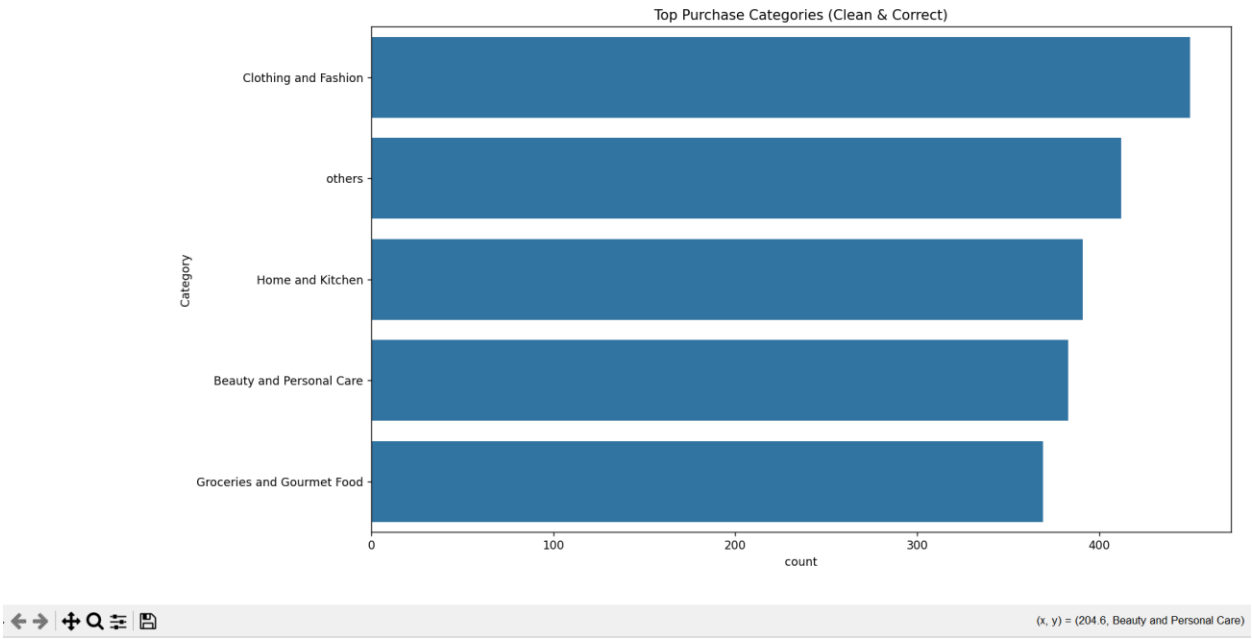
3. Data Cleaning & Preparation

- Removed duplicate records from dataset
- Handled missing values using forward fill method
- Standardized categorical data such as Gender
- Converted rating columns into numeric format
- Split multi-category values using correct delimiter (`;`)

4. Descriptive Analysis

Top Purchase Categories

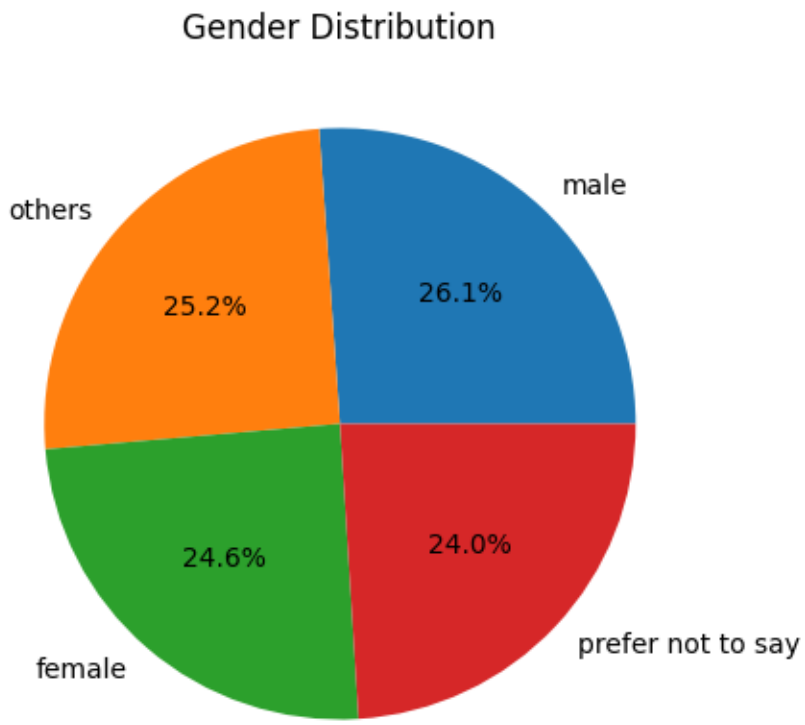
The dataset contained multiple categories in a single column. These were cleaned and split properly to get accurate category-wise frequency.



Observations:

- Clothing and Fashion is the most purchased category
- “Others” category shows high usage indicating diverse preferences
- Home and Kitchen has strong demand among customers
- Beauty and Personal Care products are frequently purchased
- Grocery purchases indicate repeated buying behavior

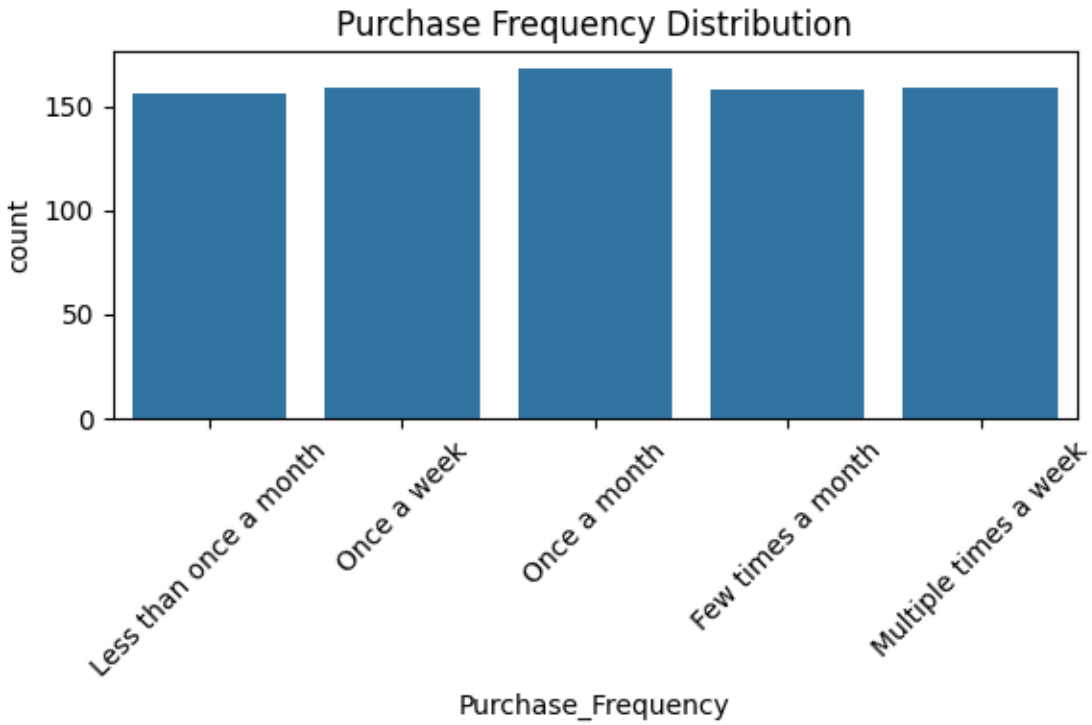
Gender Distribution Graph :



Observations:

- Customers are almost equally distributed across all genders .
- No single gender dominates the platform usage .
- Shows the platform is widely used by diverse users .
- Business strategies need not focus on one gender.

Purchase Frequency :

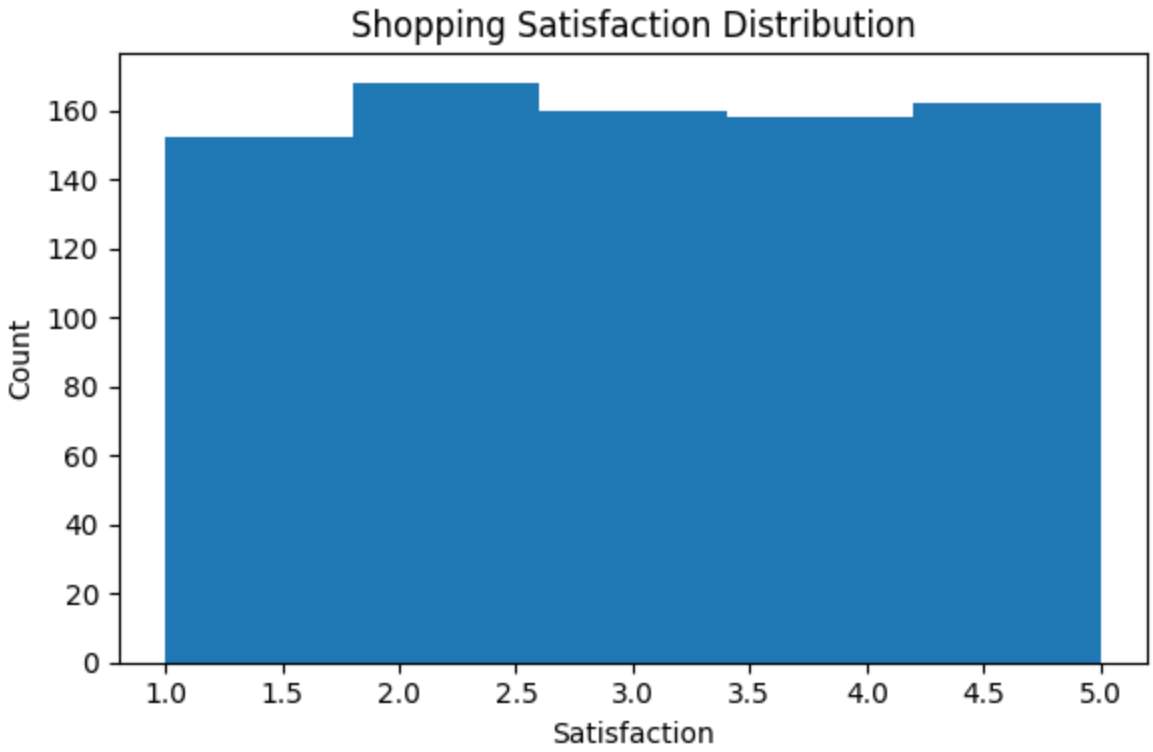


Observation :

- Majority of customers purchase regularly
- Weekly and frequent buyers are high
- Indicates strong customer engagement
- Shows good customer retention
- Platform encourages repeat purchases

Shopping Satisfaction Distribution

Customer satisfaction levels were analyzed using a histogram to understand how customers feel about their shopping experience.



Observations:

- Most customers have medium to high satisfaction levels
- Very few customers show low satisfaction
- This indicates overall positive customer experience
- However, there is still scope for improvement for low-satisfaction users

5. Customer Segmentation

K-Means clustering was applied based on:

- Shopping Satisfaction
- Customer Reviews Importance

Segments:

- **Frequent Buyers** → High satisfaction and frequent purchases
- **Occasional Shoppers** → Moderate engagement

- **At-Risk Customers** → Low satisfaction

6. Market Basket Analysis

Apriori algorithm was used to identify frequently bought product combinations.

Key Metrics:

- Support → Frequency of itemset
- Confidence → Likelihood of items bought together
- Lift → Strength of association

Findings:

- Customers tend to purchase multiple categories together
- Strong cross-selling opportunities exist
- Product combinations can improve recommendation systems

7. Key Insights

- Customers with higher satisfaction purchase more frequently
- Clothing and Fashion dominates customer preferences
- Multi-category shopping behavior is common
- Reviews play an important role in decision-making
- Daily-use items like groceries show repeat purchase patterns

8. Business Recommendations

- Improve personalized recommendation system
- Offer combo deals (e.g., clothing + accessories)
- Reduce cart abandonment by improving UI and pricing strategies
- Target low-satisfaction customers with offers
- Improve category classification to reduce “others”

Correlation Heatmap



Observations:

- Correlation value is very low (~ -0.037)
- Weak relationship between reviews and satisfaction
- Reviews do not directly impact satisfaction
- Other factors influence customer experience
- Indicates multiple variables affect buying behavior

9. Conclusion

The analysis provides valuable insights into customer behavior and purchasing patterns. These insights can help Amazon enhance customer experience, increase sales, and optimize marketing strategies.

10. Tools & Technologies Used

- Python
- Pandas, NumPy
- Matplotlib, Seaborn
- Scikit-learn (K-Means Clustering)
- Mlxtend (Apriori Algorithm)

11. Final Note

This project demonstrates how data analysis can drive business decisions and improve customer engagement.

Video Link –1 : https://www.loom.com/share/4df876b905a748838283572456956c9f

Video Link-2 :https://www.loom.com/share/fbdf5198e75b46e69afc2af6a3511f3d